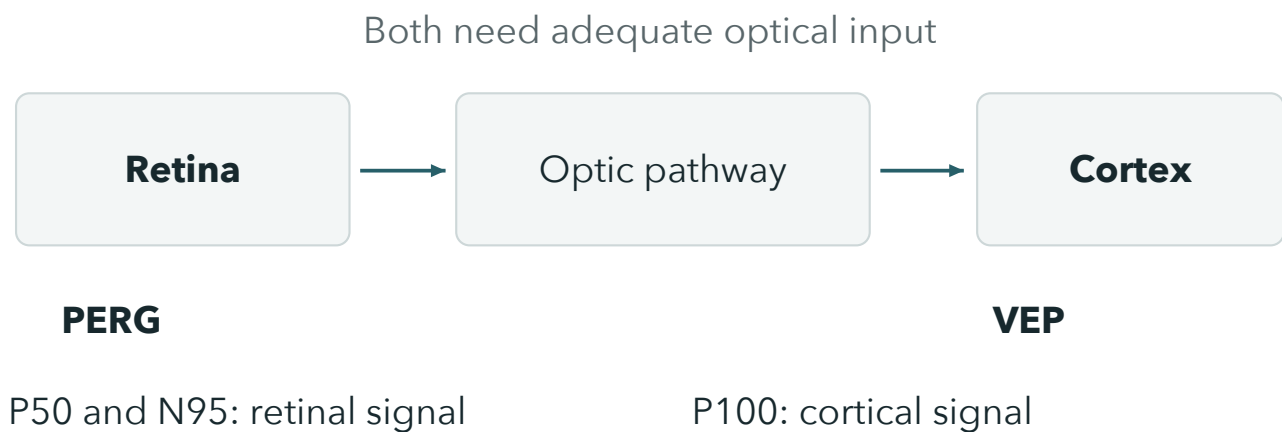


Choosing and interpreting visual electrophysiology

Part 1 | Two recordings, different generators

Retinal pattern signals and cortical pattern signals cannot be interpreted as interchangeable measurements.



Recording sites distinguish the retinal PERG from the post-retinal pathway sampled by a VEP.

A controlled dissociation

Selective N95 loss with relatively preserved P50 supports predominant ganglion-cell dysfunction. Both components still require adequate retinal input. First check refraction, media, fixation and reproducibility; an abnormal waveform is not automatically a primary neuronal lesion.

Read components before labels

A response contains amplitude and timing information; neither alone identifies every cause.

Transient PERG

Observation	Interpretation
N95 disproportionately reduced	Ganglion-cell dysfunction supported
P50 reduced	Macular input or other confounding possible
Poor repeatability	Resolve acquisition quality first

Interpret the relationship

Compare N95 with its P50 input, not only the laboratory lower limit.

A baseline problem can distort a long-latency component.

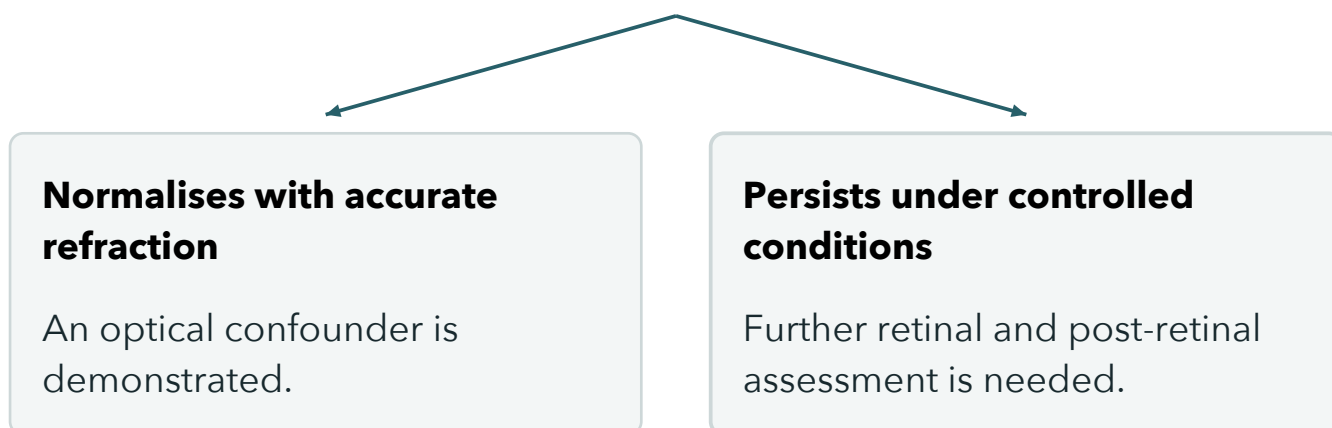
Use the same stimulus and recording conditions in longitudinal comparisons.

A VEP needs a usable retinal image

Pattern-dependent cortical responses inherit optical imperfections upstream.

Delayed pattern response

What happens after correction?



A mechanistic limit

A trial lens can improve the stimulus but cannot repair demyelinated axons within minutes. Conversely, persistence after optical correction does not by itself prove demyelination: macular dysfunction, fixation and recording conditions still require consideration.

Calibration prevents false physiology

An instrument can change the apparent response without any biological change.

Controlled quantity

Feature	Check	Why it matters
Pattern contrast	Actual display luminances	Nominal settings may mislead
Mean luminance	Stable across reversal	Unwanted luminance responses
Electrode/filter conditions	Matched protocol	Waveform and timing can change

A useful experiment

Repeat the acquisition before interpreting an isolated outlier.

Change one confounder at a time.

Use laboratory reference data obtained under compatible conditions.

Integrate, do not overclaim

Choose the recording whose generator matches the question being asked.

Clinical question

Feature	Level	Useful principle
Global retinal response	Full-field retina	Small local lesions may escape
Central retinal pattern response	Macula and ganglion cells	Optical quality is essential
Cortical pattern response	Whole afferent chain	Not an isolated nerve assay

Reasoning rehearsal

A normal downstream test does not establish that every upstream cell is normal, and an abnormal downstream test does not uniquely locate a lesion. State which structures contribute to the measured signal, then identify the additional comparison that would narrow the explanation.